25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

Matrices, Determinants & Vectors Practice Exercises

1 MCQ

Given the vectors $\mathbf{u} = \langle 3, -4 \rangle$ and $\mathbf{v} = \langle -2, 5 \rangle$, what is the vector $2\mathbf{u} + \mathbf{v}$?

A $\langle 4, -3 \rangle$

B $\langle 1, 1 \rangle$

C $\langle 4, 1 \rangle$

D $\langle 8, -13 \rangle$

2 MCQ

Matrices A and B are defined as $A = \begin{bmatrix} 2 & k \\ -1 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 4 \\ 2 & k \end{bmatrix}$, where k is a real number. If the determinant of the matrix sum $A + B$ is equal to 17, what is the value of k?

(Note: The determinant of a 2×2 matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is defined as $ad - bc$.)

A 1

B 2

C 6

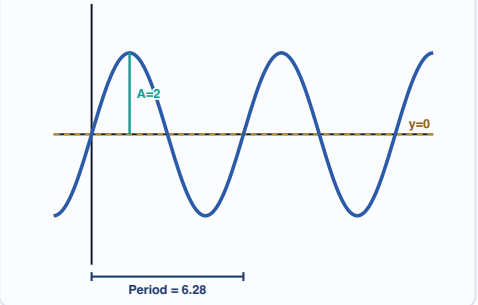
D $\frac{19}{2}$

3 MCQ

For all real values of θ, the matrix product

$$\begin{bmatrix} \cos\theta & \sin\theta \end{bmatrix} \begin{bmatrix} 6 & 2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix}$$

evaluates to a 1×1 matrix $[k]$. What is the maximum possible value of k?



A 6

B 7

C 9

D 14

4 MCQ

Given the vectors $\mathbf{u} = \langle 3, -2 \rangle$ and $\mathbf{v} = \langle 1, 5 \rangle$, what is the vector $2\mathbf{u} + \mathbf{v}$?

A $\langle 7, 1 \rangle$

B $\langle 4, 3 \rangle$

C $\langle 8, 6 \rangle$

D $\langle 5, 8 \rangle$

5 MCQ

For a 2×2 matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$, the determinant is defined as $ad - bc$. Given the matrices $A = \begin{bmatrix} x & 2 \\ -1 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -4 \\ 3 & x \end{bmatrix}$, the determinant of the matrix $A + B$ is equal to 19. If $x > 0$, what is the value of x?

A 2

B $\frac{5}{4}$

C 4

D 6

6 MCQ

Let $\mathbf{u}$ and $\mathbf{v}$ be vectors in the coordinate plane such that $|\mathbf{u}| = 5$, $|\mathbf{v}| = 10$, and the angle θ between them is obtuse ($90^\circ < \theta < 180^\circ$). If the minimum value of $|t\mathbf{u} + \mathbf{v}|$ over all real numbers t is equal to 6, what is the value of the dot product $\mathbf{u} \cdot \mathbf{v}$?

A -50

B -40

C -30

D 40

7 MCQ

What is the value of the determinant of the matrix $\begin{bmatrix} 7 & -3 \\ 2 & 4 \end{bmatrix}$?

A 34

B 22

C -29

D -34

8 MCQ

In the standard coordinate plane, vector $\mathbf{u} = \langle 2, -1 \rangle$ and vector $\mathbf{v} = \langle k, 5 \rangle$, where k is a real constant. If the magnitude of the vector $\mathbf{u} + \mathbf{v}$ is equal to 5, which of the following is a possible value of k?

A -5

B -1

C 3

D 7

9 MCQ

In the standard (x, y)-coordinate plane, vectors $\mathbf{u} = \langle a, b \rangle$ and $\mathbf{v} = \langle c, d \rangle$ have magnitudes $\|\mathbf{u}\| = 13$ and $\|\mathbf{v}\| = 5$. If their dot product is $\mathbf{u} \cdot \mathbf{v} = 25$, and the matrix $M = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ has $\det(M) > 0$, what is the value of $\det(M)$?

A 65

B 25

C 40

D 60

10 MCQ

If matrix $A = \begin{bmatrix} 3 & -2 \\ 0 & 4 \end{bmatrix}$ and matrix $B = \begin{bmatrix} -1 & 5 \\ 2 & -3 \end{bmatrix}$, what is the matrix $2A + B$?

A $\begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix}$

B $\begin{bmatrix} 5 & 3 \\ 2 & 1 \end{bmatrix}$

C $\begin{bmatrix} 5 & 1 \\ 2 & 5 \end{bmatrix}$

D $\begin{bmatrix} 7 & -9 \\ -2 & 11 \end{bmatrix}$

11 MCQ

If $A = \begin{bmatrix} -2 & 0 \\ 0 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -1 \\ 0 & 4 \end{bmatrix}$, find AB.

A $\begin{bmatrix} -2 & 1 \\ 0 & -4 \end{bmatrix}$

B $\begin{bmatrix} -2 & 4 \\ 0 & -4 \end{bmatrix}$

C $\begin{bmatrix} -1 & -1 \\ 0 & 3 \end{bmatrix}$

D $\begin{bmatrix} -2 & 2 \\ 0 & -4 \end{bmatrix}$

12 MCQ

If $A = \begin{bmatrix} 1 & -2 \\ 0 & -2 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 4 \\ 2 & 4 \end{bmatrix}$, find AB.

A $\begin{bmatrix} 4 & 2 \\ 2 & 2 \end{bmatrix}$

B $\begin{bmatrix} -1 & -4 \\ -4 & -8 \end{bmatrix}$

C $\begin{bmatrix} 3 & -14 \\ 2 & -12 \end{bmatrix}$

D $\begin{bmatrix} -1 & -2 \\ -4 & -8 \end{bmatrix}$

13 MCQ

If $A = \begin{bmatrix} -4 & -3 \\ -3 & -2 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & -4 \\ 1 & 4 \end{bmatrix}$, find AB.

A $\begin{bmatrix} 0 & -7 \\ -2 & 2 \end{bmatrix}$

B $\begin{bmatrix} -4 & -4 \\ -16 & -11 \end{bmatrix}$

C $\begin{bmatrix} -19 & 4 \\ -14 & 4 \end{bmatrix}$

D $\begin{bmatrix} -19 & 6 \\ -14 & 4 \end{bmatrix}$

14 MCQ

If $A = \begin{bmatrix} -3 & 1 \\ -3 & -5 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} -2 \\ -1 \end{bmatrix}$, find $A\mathbf{v}$.

A $\begin{bmatrix} 6 \\ 11 \end{bmatrix}$

B $\begin{bmatrix} 5 \\ 10 \end{bmatrix}$

C $\begin{bmatrix} 6 \\ 5 \end{bmatrix}$

D $\begin{bmatrix} 5 \\ 11 \end{bmatrix}$

15 MCQ

If $A = \begin{bmatrix} -5 & 0 \\ -2 & 5 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} -3 \\ 4 \end{bmatrix}$, find $A\mathbf{v}$.

A $\begin{bmatrix} 15 \\ 25 \end{bmatrix}$

B $\begin{bmatrix} 15 \\ 20 \end{bmatrix}$

C $\begin{bmatrix} 16 \\ 26 \end{bmatrix}$

D $\begin{bmatrix} 15 \\ 26 \end{bmatrix}$

16 MCQ

If $A = \begin{bmatrix} -4 & 0 \\ 3 & 1 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$, find $A\mathbf{v}$.

A $\begin{bmatrix} -12 \\ 14 \end{bmatrix}$

B $\begin{bmatrix} -11 \\ 14 \end{bmatrix}$

C $\begin{bmatrix} -12 \\ 13 \end{bmatrix}$

D $\begin{bmatrix} -12 \\ 5 \end{bmatrix}$

17 MCQ

If $A = \begin{bmatrix} -3 & -2 \\ -3 & -3 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 0 \\ -6 \end{bmatrix}$, find $A\mathbf{v}$.

A $\begin{bmatrix} 0 \\ 18 \end{bmatrix}$

B $\begin{bmatrix} 12 \\ 18 \end{bmatrix}$

C $\begin{bmatrix} 13 \\ 18 \end{bmatrix}$

D $\begin{bmatrix} 12 \\ 17 \end{bmatrix}$

18 MCQ

Suppose that $A = \begin{bmatrix} -3 & 0 \\ 1 & 5 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$, find $A\mathbf{v}$.

A $\begin{bmatrix} -15 \\ 35 \end{bmatrix}$

B $\begin{bmatrix} -14 \\ 35 \end{bmatrix}$

C $\begin{bmatrix} -15 \\ 34 \end{bmatrix}$

D $\begin{bmatrix} -15 \\ 30 \end{bmatrix}$

19 MCQ

Suppose that $A = \begin{bmatrix} -2 & -1 \\ -3 & -4 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 0 \\ -6 \end{bmatrix}$, find $A\mathbf{v}$.

A $\begin{bmatrix} 6 \\ 24 \end{bmatrix}$

B $\begin{bmatrix} 7 \\ 24 \end{bmatrix}$

C $\begin{bmatrix} 6 \\ 23 \end{bmatrix}$

D $\begin{bmatrix} 0 \\ 24 \end{bmatrix}$

20 MCQ

If $A = \begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} -1 \\ -2 \end{bmatrix}$, find $A\mathbf{v}$.

A $\begin{bmatrix} 2 \\ -2 \end{bmatrix}$

B $\begin{bmatrix} 2 \\ -3 \end{bmatrix}$

C $\begin{bmatrix} -2 \\ -4 \end{bmatrix}$

D $\begin{bmatrix} 3 \\ -2 \end{bmatrix}$

21 MCQ

If $A = \begin{bmatrix} -2 & -2 \\ -5 & 5 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} -3 \\ 0 \end{bmatrix}$, find $A\mathbf{v}$.

A $\begin{bmatrix} 6 \\ 0 \end{bmatrix}$

B $\begin{bmatrix} 7 \\ 15 \end{bmatrix}$

C $\begin{bmatrix} 6 \\ 14 \end{bmatrix}$

D $\begin{bmatrix} 6 \\ 15 \end{bmatrix}$

22 MCQ

If $A = \begin{bmatrix} 0 & -1 \\ -4 & -1 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} -1 \\ 4 \end{bmatrix}$, find $A\mathbf{v}$.

A $\begin{bmatrix} 0 \\ -4 \end{bmatrix}$

B $\begin{bmatrix} -4 \\ 0 \end{bmatrix}$

C $\begin{bmatrix} -3 \\ 0 \end{bmatrix}$

D $\begin{bmatrix} -4 \\ -1 \end{bmatrix}$

23 MCQ

If $A = \begin{bmatrix} 3 & 1 \\ 5 & 3 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} -1 \\ -6 \end{bmatrix}$, find $A\mathbf{v}$.

A $\begin{bmatrix} -8 \\ -23 \end{bmatrix}$

B $\begin{bmatrix} -9 \\ -24 \end{bmatrix}$

C $\begin{bmatrix} -9 \\ -23 \end{bmatrix}$

D $\begin{bmatrix} -3 \\ -18 \end{bmatrix}$

24 MCQ

If $A = \begin{bmatrix} -4 & -1 \\ -3 & 4 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} -2 \\ -6 \end{bmatrix}$, find $A\mathbf{v}$.

A $\begin{bmatrix} 14 \\ -19 \end{bmatrix}$

B $\begin{bmatrix} 8 \\ -24 \end{bmatrix}$

C $\begin{bmatrix} 15 \\ -18 \end{bmatrix}$

D $\begin{bmatrix} 14 \\ -18 \end{bmatrix}$

25 MCQ

Assuming that $A = \begin{bmatrix} -4 & 4 \\ 1 & 0 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$, find $A\mathbf{v}$.

A $\begin{bmatrix} 5 \\ 5 \end{bmatrix}$

B $\begin{bmatrix} 4 \\ 4 \end{bmatrix}$

C $\begin{bmatrix} -20 \\ 0 \end{bmatrix}$

D $\begin{bmatrix} 4 \\ 5 \end{bmatrix}$